Pradnya Patare

#AI, #Machine Learning, #Deep Learning, #Generative AI, #Python

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OBJECTIVE:

To be excellent AI professional and move into higher technology areas which provide an environment to improve my technical and analytical abilities.

PROFILE SUMMARY: 4+ years of IT experience as **Sr.AI/ML Engineer** , Generative AI, NLP- LLM, Data Science, Data Engineering and Computer Vision, with statistical Modeling, Machine Learning, Data Visualization skills, Deep Learning MLOps. Helped company to extract actionable insights from complex insights to both technical and non-technical audience.

EDUCATION:

- **PG-DAI (Post Graduate Diploma in Artificial Intelligence)** from **C-DAC Patna** with **66.15 %.** • **B.E.** in **Information Technology** from **SMT. Kashibai Navale College of Engineering, Pune University** with **7.16 /10 CGPA.**

SKILLS:

- **Programming Languages:** Python, C++, Java
- **Frameworks:** TensorFlow, Keras, Pytorch, NumPy, Pandas, Scikit-Learn, Langchain, LangGraph,LLM Mesh Dataiku
- **Data Visualization:** Tableau, PowerBI, Matplotlib, Seaborn, Plotly
- **ML Algorithms:** Linear, Logistic Regression, SVM, Naïve Bayes, Decision Tree, Random Forest, KNN, K-Means Clustering,
  - AdaBoost, XGBoost
- **Deep Learning:** Computer Vision, ANN, CNN, RNN, LSTM, GAN, YOLO,vLLM
- **NLP and GEN-AI:** LLM, Hugging face Language Models, BERT, GPT, Generative AI, Topic Modelling, Prompt Engineering
- **Advanced Analytics:** Regression, Classification, Clustering, Statistical Analysis
- **Databases:** MongoDB, SQL, MySQL
- **Cloud Computing:** AWS, Katonic, Docker, Kubernetes, Kafka, MLOps, GCP
- **Web Development:** Django, Flask, HTML, JavaScript, CSS
- **Experience with tools:** VS Code, PyCharm, Jupyter Notebook, Google Colab, GitHub, Git

PROJECTS:

1. AI-Based Proctoring & Video Label Identification System|Ioweb3 Technologies | 2024 - 2025

Technologies: Python, PyTorch, YOLO, OpenCV, NumPy, Pandas, Computer Vision, Deep Learning, FastAPI, Docker, Git

Project Description: Developed an AI-based video proctoring system using **YOLO and computer vision** to analyze candidate videos and identify suspicious activities through automated video labels.

Responsibilities:

- Developed **YOLO-based object detectio**n for candidate and suspicious-object identification.
- Used OpenCV for video processing, frame extraction, preprocessing, and visualization.
- Generated video labels for multiple-person presence, candidate absence, mobile/unauthorized objects, and suspicious activities.
- Implemented event analysis and risk scoring to prioritize videos for manual review.
- Evaluated and optimized model performance using precision, recall, IoU, and mAP.
- Integrated the model with APIs and supported Docker-based deployment using Git.

2. Sign Language Detection & Next Word Prediction | CDAC Project |2022

Technologies: Python, TensorFlow, Deep Learning, Computer Vision, OpenCV, CNN, NLP, TTS

Description:

- Developed a **real-time computer vision system** to detect and recognize hand signs from camera/video input using deep learning.
- Applied **image preprocessing and feature extraction** techniques to improve hand-sign recognition and classification.
- Trained a deep learning model to classify different **sign-language hand gestures** and convert recognized signs into meaningful Hindi text.
- Implemented **sequence-based Top-10 next-word prediction** to generate probable words from the detected sign sequence.
- Integrated **Text-to-Speech (TTS)** to convert the predicted Hindi text into audio for real-time communication.

- Optimized the end-to-end pipeline for **gesture detection → sign classification → text generation → speech output**, achieving approximately 90% prediction accuracy.

3. PDF Document Q&A Chatbot using RAG, Groq API and Llama3

Technology: Python, LangChain, Streamlit

Description: - Developed a chatbot for document-based question answering using a RAG approach, integrating Groq API for retrieval and Llama3 for generating responses, ensuring precise and contextually relevant answers.

WORK EXPERIENCE:

Company Name: Cyncly
Period: June 2026 – Present
Designation: Associate Manager (Sr. AI / ML Engineer)
Role: Generative AI, Data Science, Natural Language Processing Model Development, LLM's, Data Engineering, Prompt Engineering

Company Name: Persistent Systems (Ioweb3 Technologies)
Period: July 2025 – June 2026
Designation: AI / Machine Learning Engineer Consultant
Role: Generative AI, Data Science, Natural Language Processing Model Development, LLM's, Data Engineering, Prompt Engineering.

Company Name: Ioweb3 Technologies Pvt. Ltd.
Period: Jan 2024 – July 2025
Designation: Data Scientist
Role: Generative AI, Data Science, Recommendation System Development, Deep Learning, Machine Learning, Natural Language Processing Model Development, LLM's, Data Engineering.

Company Name: JetSynthesys Pvt Ltd. (Wakau Interactive Pvt. Ltd.)
Period: July 2022 – December 2023
Designation: Associate Data Engineer
Role: Data Science, Recommendation System Development, Deep Learning, Machine Learning, Natural Language Processing Model Development, Generative AI Models Development, Data Engineering.

Company Name: Extensile Pvt Ltd.
Period: March 2021 – September 2021
Designation: Data Scientist Intern
Role: Data Science, Data Collection, Deep Learning Models Development, Machine Learning Models.

COMPANY PROJECTS:
Organization: Cyncly
Technology: Python, Generative AI, Machine learning, Deep learning, NLP, Azure

Responsibilities:
- Implemented and deployed Generative AI, ML, and DL solutions using Python, GCP, AWS, Django, Docker, and Kubernetes.
- Developed AI Interview platforms, recommendation systems, chatbots, and audio/video AI applications using ML, DL, NLP, and local/self-hosted LLMs without external APIs.
- Built LangChain-based chatbots and comparison tools with RAG pipelines, embeddings, vector databases, and local LLMs for intelligent document-based question answering.
- Designed and automated end-to-end ML/AI workflows covering data preprocessing, model training, fine-tuning, deployment, monitoring, and CI/CD.
- Performed EDA and advanced analytics to identify patterns and generate actionable insights from complex datasets.

Organization: Persistent Systems (Ioweb3 Technology Pvt.Ltd.)
Technology: Python, Generative AI Models, NLP, LangChain

Responsibilities :

- Designed and developed an **LLM-powered intelligent agent** to identify **dual (duplicate) insurance claim scenarios** in the **healthcare domain**.
- Implemented automated **claim data extraction from unstructured documents** (PDFs, scanned images, text) using **Large Language Models**, generating **standardised JSON outputs**.
- Built a **duplicate claim detection framework** combining **semantic similarity (embeddings), rule-based validation, and metadata matching** (patient ID, provider, CPT/ICD codes, service dates, claim amounts).
- Integrated similarity search using **vector databases** and NLP techniques to detect exact, partial, and resubmitted duplicate claims.
- Developed **confidence-scoring and decision logic** to flag potential duplicate or fraudulent claims for downstream review. Designed and automated **end-to-end ML/LLM workflows**, including data ingestion, preprocessing, model fine-tuning, evaluation, deployment, and monitoring.
- Deployed and scaled AI solutions using **Azure, Docker, Kubernetes**, and implemented **CI/CD pipelines** for continuous model updates.

Organization: loweb3 Technologies Pvt Ltd.

Technology: Python, TensorFlow, Deep Learning Algorithms, Machine Learning Algorithms, PyTorch, Computer Vision, Django, Generative AI Models, NLP

Responsibilities:

• Implemented and deployed Generative AI, machine learning (ML), deep learning (DL) models for real-world applications using platforms like GCP, AWS, Katonic, Django, Docker, and Kubernetes.
• Developed and optimised AI models for an AI Interview platform, recommendation systems, audio- and video-based generative AI models, LLMs, Classification, Regression, Clustering, and NLP and conducted exploratory data analysis (EDA) to uncover hidden patterns and insights within complex datasets, contributing to the informed decision-making process.
• Designed advanced analytics ranging from descriptive to predictive models to machine learning techniques. • Designed and automated end-to-end ML workflows, including data collection, preprocessing, model training, fine-tuning, testing, deployment, and monitoring in production environments.
• Integrated and maintained CI/CD pipelines for seamless model deployment, version control, and continuous updates, ensuring scalable and efficient operations.

Description: Worked on multiple POC's for data collection (text, audio, image, video), data preprocessing, model training, and testing to determine the model with the best results.

Organization: JetSynthesys Pvt Ltd. (Wakau Interactive Pvt Ltd.)

Technology: Python, TensorFlow, Deep Learning Algorithms, Machine Learning Algorithms, PyTorch, Computer Vision, Django, Generative AI Models, NLP

Responsibilities:

- Implemented and deployed machine learning (ML), deep learning (DL), and generative AI models for real-world applications using platforms like Katonic, Django, Docker, and Kubernetes.
- Developed and optimised AI models for recommendation systems, audio- and video-based generative AI models, Classification, Regression, and NLP and conducted exploratory data analysis (EDA) to uncover hidden patterns and insights within complex datasets, contributing to the informed decision-making process.
- Designed advanced analytics ranging from descriptive to predictive models to machine learning techniques. • Designed and automated end-to-end ML workflows, including data collection, preprocessing, model training, fine tuning, testing, deployment, and monitoring in production environments.
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Organization: Extensile Pvt Ltd.

Technology: Python, TensorFlow, Deep Learning Algorithms, Machine Learning Algorithms, Keras, Computer Vision

Responsibilities:

- Collecting, cleaning, and preprocessing data to make it suitable for model training.
- Building and implementing machine learning and deep learning models.
- Identifying relevant features and creating new features to improve model performance.
- Designing and conducting experiments to evaluate model performance.
- Documenting the entire data science pipeline, including data preprocessing, model development, and evaluation

metrics.
- Engaging in continuous learning to stay abreast of the latest developments in the field.
- Iteratively improving models based on feedback and performance metrics.

CERTIFICATES:

- **Udemy: Complete Generative AI Course with Langchain and Huggingface**
- **Udemy: Deep Learning, Computer Vision, CNN, OpenCV, YOLO, SSD & GANs**
- **Career Essentials in Generative AI by Microsoft and LinkedIn**
- **Build Your Generative AI Productivity Skills with Microsoft and LinkedIn**
- **Introduction to Prompt Engineering for Generative AI**
- **MTA: Introduction to Programming Using JavaScript-Certified 2019 (Microsoft)**
- **NPTEL Online Certificate Course in Database Programming (IIT Kharagpur)**
- **Udemy: MySQL Database Development Mastery**